

Sprint 2 Technical Implementation Approach

Your Accessible Flight (YAF)

Proposed Technical Changes for the Chatbot

The current chatbot already has an entire pipeline for AI processing that consists of AI Handler, AI Orchestrator, Retriever interface, Staggered Failover Router, and several AI providers. In Sprint 2, we will expand this pipeline but not replace it.

The key goal here is to add at least one client-provided hotel or booking service to the existing chatbot pipeline in order for the user to get accessible accommodation data along with other travel assistance.

The proposed considerations are as follows:

- At least one hotel or booking API provided by the client should be integrated.
- The chatbot flow needs to be expanded to retrieve accommodation details prior to generating any AI replies.
- The retrieved accommodation details need to be provided to the AI Orchestrator as extra information.
- Keep using the same Staggered Failover Router to ensure reliability of the providers.
- Keep using the same /api/ai/chat endpoint to keep front-end modifications minimal.
- Work with the current architecture of AI but extend chatbot functionality.

Such modifications will allow integrating the chatbot solution with the accommodation details retrieval process.

Proposed Technical Flow

The proposed Sprint 2 workflow is shown below:



The AI Handler keeps on receiving the authenticated chat requests before sending them to the AI Orchestrator. In sprint 2, the orchestrator will get the accommodation and accessibility details that are relevant from the data sources available before forming the prompt. Once formed, the prompt is forwarded to the AI provider using the Staggered Failover Router through which the response is obtained and relayed back to the frontend.

Available Data Sources

Sprint 2 will utilise both existing internal data and new external accommodation data.

Existing Internal Data

Information regarding accessibility has been included in the PostgreSQL database, and includes the following:

- Airline accessibility information
- Airport accessibility information
- Airline equipment policies
- Battery limitations
- Airport accessibility services
- Types of equipment
- User equipment information

All this information will continue to be available for further chatbot development.

External Accommodation Data

In Sprint 2, one or more client-provided hotel or booking providers will be introduced.

The external provider can provide data like:

- Accommodations
- Accessibility
- Booking link
- Hotel data
- Availability (where applicable)

Instead of duplicating information in PostgreSQL, the chatbot will obtain the accommodation information from the external provider where needed.

Technical Requirements and Dependencies

Sprint 2 requires some already implemented components as well as new external components.

Already Implemented Components:

- Existing /api/ai/chat endpoint
- AI Handler
- AI Orchestrator
- Staggered Failover Router
- PostgreSQL database
- Authentication middleware
- Existing AI Providers (Mock, OpenAI, xAI)
- Existing retrieval support (including Pinecone integration, if enabled)

New Dependencies:

- Hotel data provided by the client, via an API
- API keys for authentication purposes
- Sandbox environment for testing
- Internet connection for accessing external services
- Backend code to retrieve accommodation information

These dependencies must be in place before the chatbot can fetch accommodation information while generating a response.

Proposed Technical Implementation Approach

Sprint 2 should be implemented by extending the current chatbot architecture in several stages.

Stage 1 – Integrate the External Data Source

Setup back-end communication using the client provided hotel or booking API.

This involves:

- configuration of API credentials
- secure API communication
- validation of returned accommodation information

Stage 2 – Retrieve Relevant Accommodation Information

When the user asks for help with a travel query, fetch data related to accommodations prior to building an AI prompt.

Examples are:

- accessible hotels
- accessible accommodations
- accessibility
- bookings
- accommodation suggestions

Stage 3 – Construct the AI Prompt

Combine:

- User's query
- Accommodation data that has been retrieved
- Accessibility data
- System instructions

into one prompt before sending it to the AI service provider.

Stage 4 – Generate the AI Response

The AI supplier uses the accommodation and accessibility information provided to create the natural language answer.

The current Staggered Failover Router still offers provider redundancy in case an AI provider is unavailable.

Stage 5 – Return the Response

The response should be sent back to the frontend through the current API endpoint /api/ai/chat.

Some possible future improvements could involve adding to the response schema such as:

- accommodation links
- hotel information
- accessibility references
- citations

Technical Considerations

External Data Accuracy

The accommodation access data is supposed to come from client-provided, verified sources whenever possible to avoid inaccurate responses.

Retrieval Before Response Generation

The relevant information related to the accommodation needs to be searched prior to the AI's response so that the response is generated based on the data available.

Minimal Architectural Changes

Sprint 2 will attempt to make use of the current AI Handler, AI Orchestrator, and Staggered Failover Router whenever possible to reduce implementation difficulty.

External API Reliability

The hotel API or even the booking API may undergo downtime or higher response times. In such an event, the chatbot should still run without any issues.

Authentication and Security

The external API keys have to be kept securely with environment variables and must never be written into the code.

Future Extensibility

The solution will be designed in a way that will allow new hotel providers or travel-related API integration in future sprints without major changes to architecture.

Summary

Sprint 2 will extend the current YAF chatbot implementation in that it will allow the inclusion of client-provided accommodation and booking details within the current AI processing flow. This will involve using the existing AI Handler, AI Orchestrator, Staggered Failover Router and /api/ai/chat API while adding accommodation detail retrieval prior to AI response generation.

This implementation strategy allows the user to get accessibility related accommodation detail from external data sources approved by YAF while retaining the existing backend architecture of YAF. This helps reduce changes in the architecture as well as define the necessary dependencies for Sprint 2 implementation.